

OLAP (Online Analytical Processing)

Introduction

- OLAP is a technology that allows users to perform multidimensional analysis of data in data warehouses.
- It supports complex queries, trend analysis, and decision support.
- OLAP systems are categorized into ROLAP, MOLAP, and HOLAP.

1. Types of OLAP

a) ROLAP (Relational OLAP)

- Data stored in relational databases (tables).
- Uses SQL queries to perform analysis.
- Suitable for very large data warehouses.

Advantages:

- Handles large volumes of data.
- Leverages existing relational DBMS.

- Good scalability.

Disadvantages:

- Slower performance due to SQL joins.
- Limited by relational model constraints.

b) MOLAP (Multidimensional OLAP)

- Data stored in multidimensional cubes.
- Pre-computed aggregations → faster query response.

Advantages:

- Very fast query performance.
- Efficient for complex calculations.
- Intuitive cube representation.

Disadvantages:

- Limited storage capacity (not suitable for huge data).
- Pre-computation may take time.

c) HOLAP (Hybrid OLAP)

- Combines ROLAP + MOLAP.

- Detailed data stored in relational DB (ROLAP).
- Aggregated data stored in multidimensional cube (MOLAP).

Advantages:

- Balance between storage and speed.
- Can handle large volumes while keeping performance good.

Disadvantages:

- More complex architecture.
- Higher implementation cost.

2. OLAP Operations

1. Roll-up → Summarize data (e.g., day → month → year).

2. Drill-down → Go into finer details (e.g., year → month → day).

3. Slice → Select a single dimension (e.g., sales in 2023).

4. Dice → Select multiple dimensions (e.g., sales of TVs in 2023 in India).

5. Pivot (Rotate) → Reorient cube for different perspective (e.g., sales by product vs. sales by region).

3. Advantages of OLAP

- Multidimensional data view.
- Fast query performance (especially MOLAP).
- Supports complex calculations and trend analysis.
- User-friendly for decision-making.
- Flexible operations (slice, dice, roll-up, drill-down).

4. Disadvantages of OLAP

- ROLAP: Slow for complex queries.
- MOLAP: Storage limitations.
- HOLAP: Expensive and complex.
- Requires large preprocessing time.
- Not suitable for real-time transactional data.

Conclusion

- ROLAP → best for large data, scalable but slower.
- MOLAP → best for speed, but limited storage.
- HOLAP → compromise between both, but complex.
- OLAP systems play a vital role in data warehousing and decision support by enabling fast, multidimensional analysis.